1. Software Test Guide

HYBRID_LIB_ADA

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Author(s)	Michael Gardner
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Project Profile	
Variant	library
Library Role	enterprise
Application Role	N/A
Assurance	spark-targeted
Execution	sequential
Parallelism	none
Platform	desktop, server, embedded
Execution Environment	linux, macos, windows, ada-runtime
Processor Architecture	amd64, arm64, arm32
Deployment	native

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2. Introduction

2.1. Purpose

This Software Test Guide (STG) describes the testing strategy, test structure, execution procedures, and maintenance guidance for **HYBRID_LIB_ADA**.

2.2. Scope

This document covers:

- test architecture and organization
- unit and integration test suites
- test execution procedures
- shared test framework usage
- adding new tests
- runnable example programs

2.3. References

- Software Requirements Specification (SRS)
- Software Design Specification (SDS)
- `docs/guides/all_about_our_api.md`

3. Test Strategy

3.1. Test Categories

Category	Location	Purpose
Unit Tests	`test/unit/`	Test individual packages in isolation.
Integration Tests	`test/integration/`	Test cross-layer interactions.
Example Programs	`examples/`	Runnable demonstrations of library usage.

See CHANGELOG for current test counts per release.

3.2. Testing Philosophy

- **Result Monad Testing:** all error paths are tested via Result inspection.
- **Mock-Based Isolation:** infrastructure is mocked for unit tests where appropriate.
- **No Exceptions for Expected Failures:** tests verify Result-based expected failure handling.
- **Deterministic:** same inputs produce the same observable results.
- **Public API Coverage:** all public APIs should be covered by tests.

4. Test Organization

4.1. Directory Structure

```
test/
├── bin/                        # Compiled test executables
│   ├── unit_runner
│   ├── integration_runner
│   └── test_*.adb executables
│
├── common/                     # Shared test infrastructure
│   ├── test_framework.ads
│   └── test_framework.adb
│
```

```
├── unit/                          # Unit test sources
│   ├── unit_tests.gpr
│   ├── unit_runner.adb
│   ├── test_domain_error_result.adb
│   ├── test_domain_person.adb
│   ├── test_application_command_greet.adb
│   ├── test_application_usecase_greet.adb
│   └── test_api_operations.adb
│
├── integration/                   # Integration test sources
│   ├── integration_tests.gpr
│   ├── integration_runner.adb
│   └── test_api_greet.adb
│
└── python/                        # Python-based tests
    └── test_arch_guard_ada.py
```

4.2. Naming Conventions

Element	Convention	Example
Test file	`test_<layer>_<package>.adb`	`test_domain_person.adb`
Test name	Descriptive, action-result	`"Create valid name - Is_Ok"`
Mock prefix	`Mock_`	`Mock_Writer_Success`
Runner	`<category>_runner.adb`	`unit_runner.adb`

4.3. GPR Projects

Tests use `hybrid_lib_ada_internal.gpr`, which provides unrestricted access to project packages for testing without `Library_Interface` restrictions.

5. Test Framework

5.1. Framework Overview

The shared test framework in `test/common/test_framework.ads` provides:

- test result tracking
- category summaries
- grand total reporting
- color-coded output

5.2. Framework API

```
package Test_Framework is

   procedure Register_Results (Total : Natural; Passed : Natural);

   function Grand_Total_Tests return Natural;
   function Grand_Total_Passed return Natural;

   procedure Reset;

   function Print_Category_Summary
     (Category_Name : String;
      Total         : Natural;
      Passed        : Natural) return Integer;
```

```
end Test_Framework;
```

5.3. Usage Pattern

```
procedure Test_My_Package is
   Total_Tests  : Natural := 0;
   Passed_Tests : Natural := 0;

   procedure Run_Test (Name : String; Passed : Boolean) is
   begin
      Total_Tests := Total_Tests + 1;
      if Passed then
         Passed_Tests := Passed_Tests + 1;
      end if;
   end Run_Test;

begin
   --  Test cases here
   Run_Test ("Feature works", Condition);

   Test_Framework.Register_Results (Total_Tests, Passed_Tests);
end Test_My_Package;
```

6. Test Execution

6.1. Running All Tests

```
make test-all
```

6.2. Running Specific Suites

```
make test-unit
make test-integration
```

6.3. Expected Output

```
========================================
     HYBRID_LIB_ADA UNIT TEST SUITE
========================================

[PASS] ...
...
```

Note: current exact counts are maintained in CHANGELOG.

7. Test Details

7.1. Unit Tests

7.1.1. Domain Layer Tests

- `test_domain_error_result.adb` — tests `Domain.Error.Result`
- `test_domain_person.adb` — tests `Domain.Value_Object.Person`
- `test_domain_option.adb` — tests `Domain.Value_Object.Option`

7.1.2. Application Layer Tests

- `test_application_command_greet.adb` — tests `Application.Command.Greet`

- `test_application_usecase_greet.adb` — tests `Application.Usecase.Greet` with mock writer isolation

7.1.3. API Layer Tests

- `test_api_operations.adb` — tests `Hybrid_Lib_Ada.API.Operations`
- verifies SPARK-safe generic instantiation

7.2. Integration Tests

- `test_api_greet.adb` — tests full stack through `Hybrid_Lib_Ada.API`
- uses the real console writer adapter
- verifies complete workflow end-to-end

7.3. Architecture Validation Tests

`test/python/test_arch_guard_ada.py` validates direct dependency rules for the library architecture profile. This validation is run during normal developer workflow and again in CI/CD.

8. Writing New Tests

8.1. Test Template

```
pragma Ada_2022;
with Ada.Text_IO; use Ada.Text_IO;
with Test_Framework;
with My.Package;

procedure Test_My_Package is
   Total_Tests  : Natural := 0;
   Passed_Tests : Natural := 0;

   procedure Run_Test (Name : String; Passed : Boolean) is
   begin
      Total_Tests := Total_Tests + 1;
      if Passed then
         Passed_Tests := Passed_Tests + 1;
      end if;
   end Run_Test;

begin
   Run_Test ("Feature works", My.Package.Feature = Expected);
   Test_Framework.Register_Results (Total_Tests, Passed_Tests);
end Test_My_Package;
```

8.2. Adding to GPR

Update the relevant test project file to include the new test executable in `for Main use (...)`.

8.3. Mock Patterns

Use simple, resettable mock state for outbound-port behavior:

- captured message
- call count
- configured success/failure mode

9. Traceability

9.1. Requirements to Tests

Requirement	Test File
FR-01.1 (Person)	`test_domain_person.adb`
FR-01.4 (Error)	`test_domain_error_result.adb`
FR-02.2 (Use Case)	`test_application_usecase_greet.adb`
FR-04.4 (Operations)	`test_api_operations.adb`
FR-04.1 (Facade)	`test_api_greet.adb`

9.2. Layer Coverage

Layer	Test Files
Domain	`test_domain_*.adb`
Application	`test_application_*.adb`
API	`test_api_*.adb`

See CHANGELOG for current counts per layer and total.

10. Example Programs

10.1. Basic Greeting

- file: `examples/basic_greeting.adb`
- purpose: demonstrate the simplest library usage path
- build: `alr exec -- gprbuild -P examples/examples.gpr`
- run: `./examples/bin/basic_greeting`

10.2. Error Handling

- file: `examples/error_handling.adb`
- purpose: demonstrate Result-based validation and error handling
- run: `./examples/bin/error_handling`

10.3. Building Examples

Examples are built via `examples/examples.gpr`.

11. Test Maintenance

11.1. When to Update

- new package added → add corresponding test file
- API changed → update affected tests
- bug fixed → add regression test
- error handling changed → update error-path tests

11.2. Quality Guidelines

- test names must be descriptive
- one assertion per test case is preferred
- mock state is reset between tests
- tests shall not depend on one another

11.3. Continuous Integration

Tests run automatically on:

- push to main branch
- pull request creation
- manual workflow dispatch

11.4. Exit Codes

Code	Meaning
0	All tests passed
1	One or more tests failed

12. Appendices

12.1. Change History

Version	Date	Author	Changes
2.1.0	2025-12-14	Michael Gardner	Remove hardcoded metrics per documentation standards; metrics now in CHANGELOG.
2.0.0	2025-12-09	Michael Gardner	Complete regeneration for v2.0.0; added Example Programs section.
1.0.0	2025-11-29	Michael Gardner	Initial release.